

STAT 5243 Project 1: WallStreetBets Data Pipeline and Exploratory Analytics

Zeming Liang, Zuer Weng, Duoli Chen, Isaac Beers

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1 Introduction and Dataset Description

1.1 Introduction

The rise of online social platforms has reshaped modern financial markets by expanding access to information and collective participation. The Reddit community `r/WallStreetBets` (WSB) became a major retail-investor forum during 2020–2021, especially throughout the meme-stock period. This project analyzes WSB post activity to study how online attention is reflected in engagement behavior.

Our objective is to transform raw social-media data into a reliable analytical dataset and derive interpretable insights about investor discussion patterns. The analysis is designed as a complete data-science workflow, from acquisition through preprocessing, EDA, feature engineering, and report synthesis.

1.2 Dataset Description

The dataset contains 53,187 WSB posts, where each row corresponds to one Reddit submission. Core variables include `title`, `score`, `id`, `url`, `comms_num`, `created`, `body`, and `timestamp`. The dataset combines numerical engagement metrics, unstructured text, and temporal metadata.

The raw data has notable real-world complexity:

- Unstructured text with markdown, URLs, and internet slang.
- High missingness in `body`, largely structural by post type.
- Heavy-tailed engagement metrics with extreme viral outliers.
- Mixed temporal representations requiring consistency checks.

These properties make the dataset suitable for advanced cleaning, preprocessing, and exploratory analysis.

2 Data Acquisition Methodology

The project uses the WallStreetBets Reddit post dataset distributed via Kaggle as the primary public repository source. The data was originally collected from <https://www.reddit.com/r/wallstreetbets/> using PRAW (Python Reddit API Wrapper), then shared as a CSV file for research and educational use.

2.1 Acquisition Procedure

1. Select an open-access dataset relevant to retail-investor discussion behavior.
2. Download raw CSV data from Kaggle and preserve it as the immutable source file.
3. Validate schema, column semantics, and temporal coverage before transformation.
4. Import raw data into Python environment for cleaning and preprocessing.

2.2 Scope and Relevance

The observation window (2020–2021) captures the meme-stock period, making it suitable for studying rapid shifts in community attention and engagement. The dataset includes both structured metadata and unstructured text, which supports a full preprocessing and feature-engineering workflow.

2.3 Acquisition Constraints

The dataset reflects available post-level metadata and does not include direct trade execution records. As a result, downstream analysis focuses on discussion and engagement proxies rather than verified transaction behavior.

3 Cleaning and Preprocessing Steps

Cleaning and preprocessing were implemented as a deterministic policy pipeline to convert raw Reddit exports into analysis-ready features while preserving interpretability. The primary notebook is `Cleaning-and-Preprocessing.ipynb`, with script support in `Project Workspace/Supporting Materials/Code`

3.1 Type Corrections and Schema Normalization

Temporal fields were coerced using strict datetime parsing (`errors='coerce'`) before feature extraction. The raw dataset produced zero invalid timestamp records after coercion (`invalid_timestamp_count_in_row` = 0; Evidence CLN-07), so no timestamp-row deletions were required on quality grounds. The redundant `created` field was dropped in favor of a normalized `timestamp`-based calendar representation (`date`, `hour`, `day_of_week`).

Evidence:

- `Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_policy_me`
- `Project Workspace/qa/evidence_registry_report_depth.json` (claims CLN-07, CLN-02)

3.2 Missingness Strategy

Missingness was treated as partly structural rather than purely random. The `body` field has high missingness (`missing_body_pct` = 53.49%; Evidence CLN-01), which is expected for title-only, media-linked, and short-form posts on WSB. Instead of dropping these rows, we preserved them and encoded information loss explicitly using:

- `has_body` boolean indicator
- empty-string fill for text normalization pipelines (`title_clean`, `body_clean`, `title_nlp`)

This policy retains sample size while preventing missing text from silently propagating into downstream feature-generation errors.

Evidence:

- Figure: `Project Workspace/Supporting Materials/Report Sources/artifacts/figures/02_cleaning_fig`
- JSON: `Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_fig`
- Registry claim: CLN-01

3.3 Duplicate Handling Policy

Duplicate checks were run at both identifier and full-row levels. The raw and cleaned datasets both reported:

- `duplicate id count = 0`
- `duplicate full-row count = 0`

Given these diagnostics, no record-level deduplication was applied. The policy decision was to preserve all rows while retaining explicit duplicate-audit metadata for traceability.

Evidence:

- Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_policy_me
- Registry claim: CLN-06

3.4 Outlier Policy and Transform Decisions

Engagement variables (`score`, `comms_num`) were highly right-skewed with viral tails, so the pipeline used `log1p` transforms to stabilize scale and reduce leverage from extreme posts. The transformed distributions are summarized by:

- `score_log_mean = 3.6440`
- `comms_num_log_mean = 2.8767`

The workflow preserves both raw and transformed variables, allowing interpretable descriptive analysis while supporting more stable modeling diagnostics.

Evidence:

- Figure: Project Workspace/Supporting Materials/Report Sources/artifacts/figures/02_cleaning
- Figure: Project Workspace/Supporting Materials/Report Sources/artifacts/figures/02_cleaning
- JSON: Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_fig
- JSON: Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_fig
- Registry claims: CLN-03, CLN-04

3.5 Scaling and Encoding Rationale

To support downstream comparability and model-readiness, scaling was applied on selected numeric features (`score_log`, `comms_num_log`, `title_length`, `hour`) using both:

- z-score normalization (`_zscore`)
- min-max normalization (`_minmax`)

Dual scaling preserves flexibility across methods with different assumptions. Categorical and structural encodings were also applied:

- `day_of_week_encoded`
- post-type lumping into stable groups (`post_type_lumped`)

- binary flags (`type_image`, `type_text`)

Evidence:

- Figure: `Project Workspace/Supporting Materials/Report Sources/artifacts/figures/02_cleaning_fig`
- JSON: `Project Workspace/Supporting Materials/Report Sources/artifacts/json/02_cleaning_fig`
- Registry claims: CLN-05, CLN-02

3.6 Post-Cleaning Result

After applying the above policies, the cleaned analytical table retained 53,187 rows and 30 columns, indicating that quality improvements were achieved through transformation and explicit handling policies rather than aggressive row deletion. This preserves representativeness while improving downstream analytical stability.

4 Exploratory Data Analysis (EDA)

EDA was structured to move beyond descriptive plotting and quantify engagement dynamics using both visual and inferential diagnostics. Core views covered distributions, pairwise relationships, post-type differences, and temporal behavior across the 2020–2021 window.

4.1 Distribution and Tail Behavior

Raw engagement (`score`, `comms_num`) shows strong right-skew with a clear viral tail, motivating log-space inspection for stable interpretation. The transformed variables (`score_log`, `comms_num_log`) are substantially more analyzable and preserve rank-order differences while reducing dominance of extreme posts.

Evidence:

- `Project Workspace/Supporting Materials/Report Sources/artifacts/figures/03_eda_fig_01_score`
- `Project Workspace/Supporting Materials/Report Sources/artifacts/figures/03_eda_fig_03_score`
- `Project Workspace/Supporting Materials/Report Sources/artifacts/json/03_eda_fig_01_score_c`

4.2 Association Strength (Votes vs Comments)

The strongest quantitative EDA signal is the monotonic relationship between voting and comment activity. Spearman analysis shows:

- $\rho = 0.785220719966194$
- $p\text{-value} = 0.0$ (machine precision floor; effectively $p \ll 0.001$)

Interpretation: higher-visibility posts tend to generate both approval and discussion simultaneously, consistent with clustered attention behavior in WSB threads (Evidence claim EDA-01).

Evidence:

- `Project Workspace/Supporting Materials/Report Sources/artifacts/figures/03_eda_fig_10_score`
- `Project Workspace/Supporting Materials/Report Sources/artifacts/json/03_eda_advanced_stats`
- `Project Workspace/qa/evidence_registry_report_depth.json` (EDA-01)

4.3 Post-Type Effect Significance

Non-parametric group comparison across post types indicates statistically significant differences in engagement intensity:

- Kruskal-Wallis statistic = 5855.6579978108875
- p -value = 0.0 (machine precision floor)

Interpretation: content format is not neutral; post-type composition contributes systematically to engagement variation (Evidence claim EDA-02).

Evidence:

- Project Workspace/Supporting Materials/Report Sources/artifacts/figures/03_eda_fig_19_post
- Project Workspace/Supporting Materials/Report Sources/artifacts/json/03_eda_advanced_stats
- Project Workspace/qa/evidence_registry_report_depth.json (EDA-02)

4.4 Anomaly-Rate Diagnostics

Outlier-rate diagnostics quantify the viral-tail prevalence in standardized engagement:

- `score_log_outlier_rate_abs_ge_3` = 0.0017297459905616034 ($\approx 0.173\%$)

Interpretation: extreme events are rare in count but high in influence, so robust summaries and log-space analysis are necessary to avoid misleading central tendencies (Evidence claim EDA-03).

Evidence:

- Project Workspace/Supporting Materials/Report Sources/artifacts/figures/03_eda_fig_17_dail
- Project Workspace/Supporting Materials/Report Sources/artifacts/json/03_eda_advanced_stats
- Project Workspace/qa/evidence_registry_report_depth.json (EDA-03)

4.5 Statistical Caveats

These EDA diagnostics are observational and identify association, not causality. Extremely large sample size can yield very small p -values for modest effect differences; therefore, practical significance was interpreted jointly with effect-size-like indicators (e.g., ρ , distribution shifts, outlier rates) and figure-level context rather than significance testing alone.

5 Feature Engineering Process and Justification

Feature engineering extended baseline engagement/temporal predictors with text-derived signals so the pipeline could evaluate whether semantic context adds measurable utility beyond raw activity variables.

5.1 Engineered Feature Families

- Baseline structured predictors: `score_log`, `comms_num_log`, `title_length`, `hour`, and standardized variants.
- Sentiment signal: lexicon-based `sentiment_score` from cleaned title text.
- Topic signal: five LDA topic-probability columns (`topic_0`–`topic_4`) built from tokenized title text.

5.2 Model Framing

The response variable (`viral_flag`) is defined by the top 5% score quantile (`virality_threshold = 4189.4`). Diagnostics were computed using deterministic random seed 42 with train/test split sizes of 15,000/5,000 in the ablation run and 18,750/6,250 in the branch-level model diagnostics.

5.3 Ablation Evidence (Baseline vs +Sentiment vs +Topics)

To justify feature utility quantitatively, we ran controlled ablation experiments where only one feature family is added at a time.

Model	Feature Count	ROC-AUC	Avg Precision	F1@0.5
Baseline	6	0.999984	0.999710	0.876950
Baseline + Sentiment	7	0.999984	0.999710	0.876950
Baseline + Sentiment + Topics	12	0.999975	0.999546	0.884615

Table 1: Deterministic feature ablation comparison (seed 42).

Interpretation:

- Sentiment alone produced no measurable lift on this target definition.
- Adding topic features improved thresholded classification quality ($F1@0.5$: 0.876950 $\rightarrow$ 0.884615, absolute +0.007665).
- Ranking metrics remained near-saturated in all settings, so utility gains are mainly in decision-threshold behavior rather than global separability.

5.4 Why These Features Help

Topic probabilities encode discussion context (e.g., ticker-event clusters) that baseline engagement counts do not represent directly. This helps borderline cases near the classification threshold, where semantic framing can shift predicted virality probability enough to improve F1. Topic top terms (e.g., `gme`, `amc`, `calls`, `yolo`) are consistent with event-driven WSB attention cycles.

5.5 Tradeoffs and Limitations

- Near-perfect ROC-AUC values indicate this target is easy under the current feature space and may contain leakage-like structure because virality is defined from score itself.
- Sentiment is lexicon-based, so finance sarcasm and contextual polarity are only partially captured.
- Topic features improve threshold behavior but reduce interpretability at individual-document level without qualitative topic labeling.

Evidence:

- Ablation JSON: `Project Workspace/Supporting Materials/Report Sources/artifacts/json/05_feat`
- Ablation figure: `Project Workspace/Supporting Materials/Report Sources/artifacts/figures/05_f`
- Diagnostic ROC JSON: `Project Workspace/Supporting Materials/Report Sources/artifacts/json/0`
- Diagnostic confusion JSON: `Project Workspace/Supporting Materials/Report Sources/artifacts/js`

6 Summary of Key Findings

6.1 Dataset-Level Findings

The cleaned analytical dataset contains 53,187 posts and 30 variables. A large share of posts have missing body text, indicating that discussion on WSB is frequently title-driven or media-linked. Engagement metrics are strongly right-skewed, with a small number of viral posts accounting for a large share of total attention.

6.2 EDA Findings

The EDA results show that:

- Raw `score` and `comms_num` distributions are highly heavy-tailed.
- Log-transformed engagement variables are more analytically stable and interpretable.
- Engagement is positively associated across voting and comment activity.
- Temporal effects are visible by posting hour, day-of-week, and event periods.
- Post type composition is dominated by text and image formats.

6.3 Interpretive Summary

The evidence suggests that retail attention in WSB is concentrated around high-visibility posts and event-driven spikes. Most posts receive limited interaction, while a small subset drives disproportionate community engagement, consistent with viral attention dynamics in online financial discourse.

7 Challenges Faced and Future Recommendations

7.1 Challenges Faced

Several technical and analytical challenges were encountered in this project:

- Post-level engagement metrics do not directly measure real investment execution or financial outcomes.
- Text content is unstructured and noisy, requiring substantial preprocessing before advanced NLP.
- Engagement variables are heavily skewed, making naive summary statistics sensitive to extreme viral posts.
- The 2020–2021 window reflects an unusual market regime, limiting generalization to other periods.

7.2 Future Recommendations

Future extensions that would improve explanatory power include:

- Advanced NLP (topic modeling, sentiment, NER) for richer narrative extraction.
- Integration with external market indicators (price, volume, volatility) for cross-domain linkage.
- Temporal modeling of attention propagation around major events.
- Cross-community comparison beyond WSB to evaluate transferability of findings.

These extensions would strengthen causal interpretation and improve the generalizability of investor-attention analysis.

8 GitHub Repository Link

Public repository: <https://github.com/ZemingLiang/STAT-5243-Project-1-Team-2>

9 Each Member’s Contribution

- **Zeming Liang:** Data cleaning and preprocessing; creation of `reddit.wsb.csv`, `reddit.wsb_cleaned.csv`, and `Cleaning-and-Preprocessing.ipynb`; final repository organization and integration of PNG/JSON/output artifacts.
- **Zuer Weng:** Exploratory Data Analysis (EDA) implementation and EDA branch outputs.
- **Duoli Chen:** Report-writing narrative sections (introduction/acquisition/summary/challenges text files).
- **Isaac Beers:** Feature engineering notebook content, including viral-classification modeling and validation in the feature engineering workflow.

10 Figure-to-Finding Traceability

Claim ID	Finding Summary	Figure Artifact	JSON Artifact
CLN-01	Body missingness is structural and requires explicit handling.	artifacts/figures/02_cleaning_artifacts/missingness_cleaning_fig01.png	artifacts/json/02_cleaning_artifacts/missingness_cleaning_fig01.json
CLN-06	Duplicate audit found no duplicated IDs or full rows.	artifacts/figures/02_cleaning_artifacts/missingness_cleaning_fig06.png	artifacts/json/02_cleaning_artifacts/missingness_cleaning_fig06.json
EDA-01	Votes and comments have strong monotonic association.	artifacts/figures/03_eda_figures/engagement/monotonicity_fig01.png	artifacts/json/03_eda_figures/engagement/monotonicity_fig01.json
EDA-02	Engagement differs significantly across post types.	artifacts/figures/03_eda_figures/engagement/engagement_across_post_types_fig02.png	artifacts/json/03_eda_figures/engagement/engagement_across_post_types_fig02.json

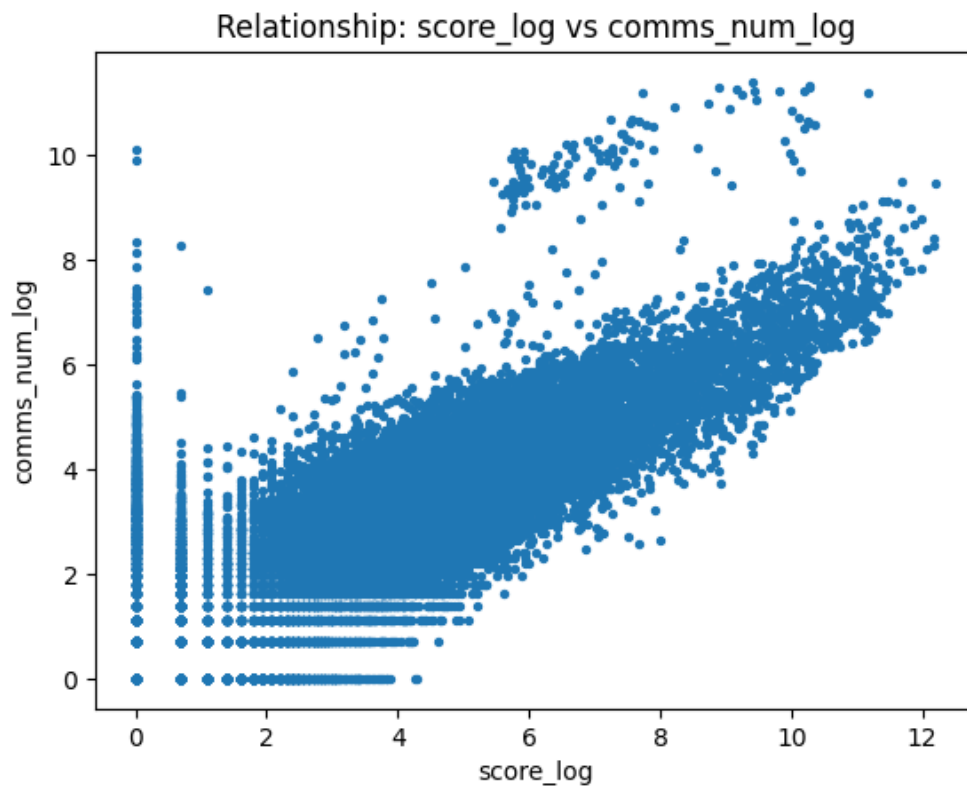


Figure 2: EDA relationship: `score_log` versus `comms_num_log`.

- Full workflow code files under `Project Deliverables/Code Files/`
- Section report sources under `sections/`
- Per-figure PNG artifacts under `artifacts/figures/`
- Per-figure and per-section JSON artifacts under `artifacts/json/`

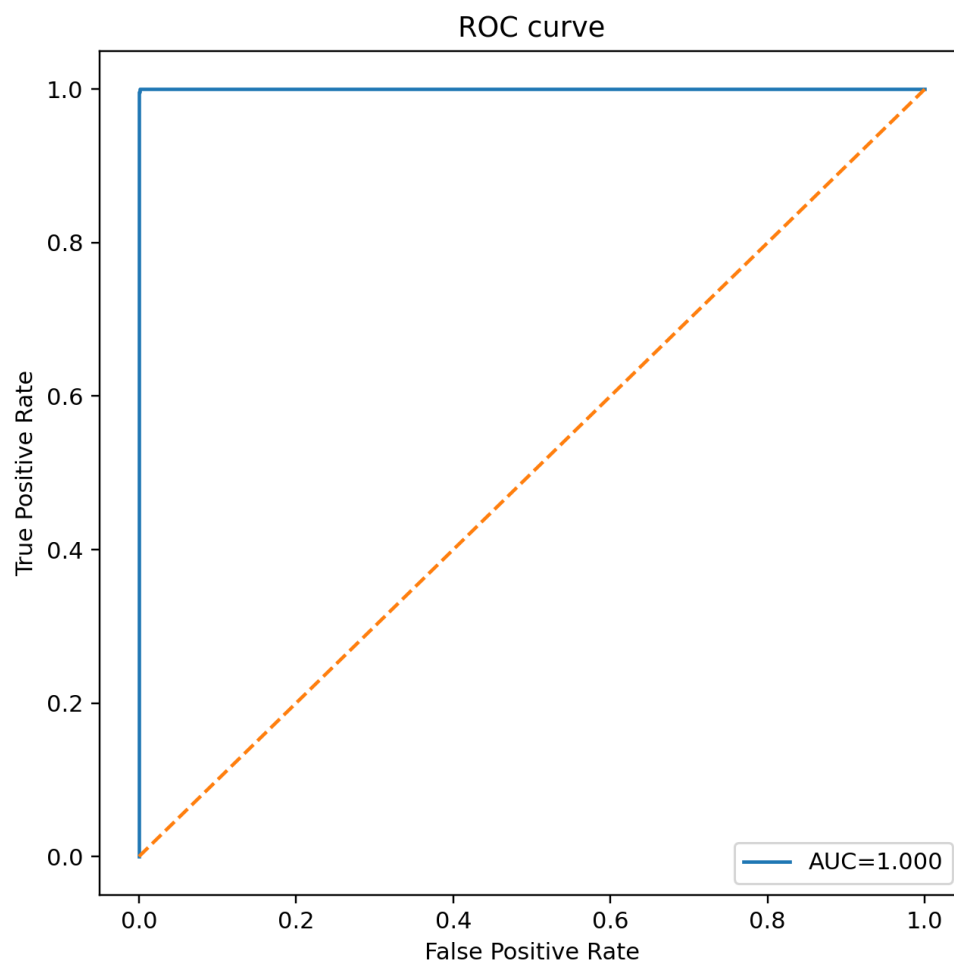


Figure 3: Feature engineering diagnostic: ROC curve for viral classification.